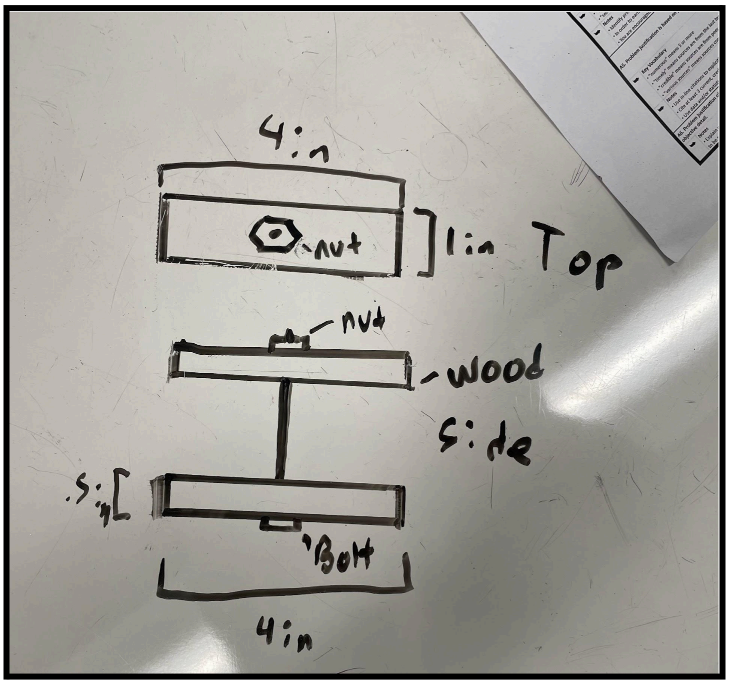
Element D

We decided to begin the brainstorming process by individually coming up with ideas and sharing them collectively as a group. This way, we can broaden the amount of ideas we have while creating the initial design and decision matrices.

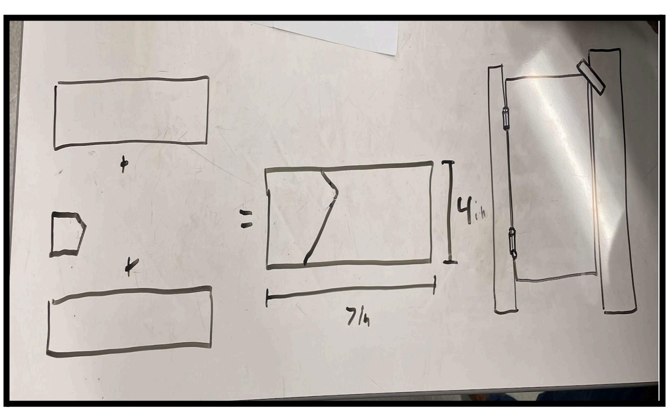
Student A:



- Opting for a clamp design. Slips in between the crack in the door and clamps, screwing in and allowing no rotation.

Explanation:
Since the design needed to be operable outside of the bathroom stall, a mechanism outside of the door needed to be in place. Student A derived this design from various types of clamps which he thought would be applicable to this project.

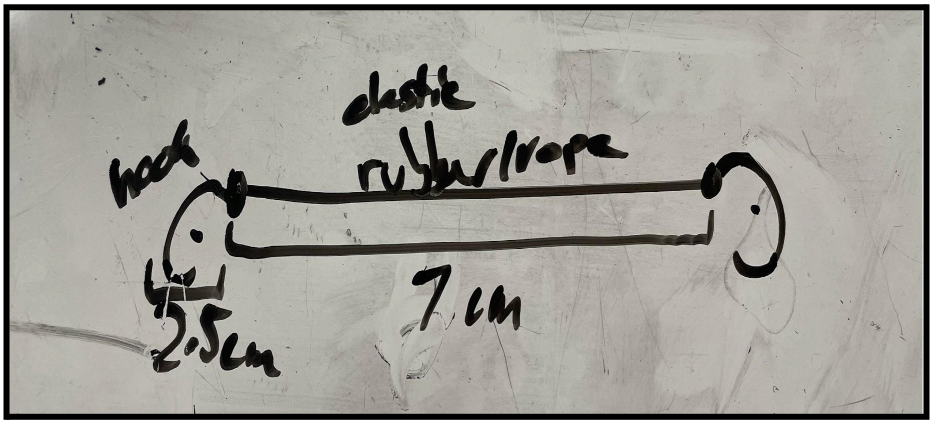
Student B:



-A wooden block that slips over the stall door and the stall divider, allowing no rotation.

Explanation: Student B designed this based around the top three criteria. The amount the door can rotate, the time the lock can hold the door closed, and the force that the lock can withstand. Designing a lock that had strengths in these categories, he developed this.

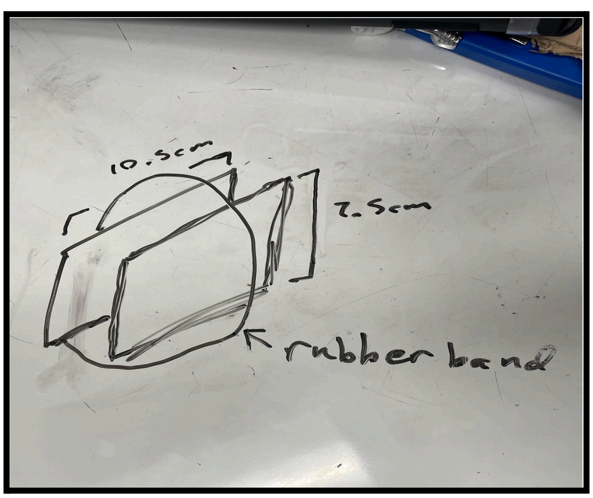
Student C:



- Two hooks with elastic bands that latch onto the door and the divider, allowing for minimal rotation.

Explanation: Student C designed this lock around the comfort and portability aspects of the criteria, designing a highly portable, lightweight, and flexible design that is easily carried in the primary stakeholders pocket.

Student D:



- Two metal plates sandwich both the door and the divider, locking it in place. The rubber band wrapped around the plates holds the two rubber plates against the door.

Explanation: Similar to student C's design, student D designed this based

Design Matricy:

Criteria	Importance (%)	Design A	Design B	Design C	Design D
Door Swing	40%	9	9	2	9
Force Against the lock	25%	7	8	5	8
Time holding door closed	15%	9	10	7	7
Fits in a pocket	10%	6	2	10	10
2-3 lbs	6%	6	7	9	8
Occupied Label	4%	3	9	1	3
Total	100%	7.78	8.08	4.68	8.15

Note that importance weightings were previously decided in Element C.

Analysis:
Design D scored the highest by a small margin, with design B following closely behind. Design D scored very high in the door swing category along with designs A and B. Meaning it would not likely allow much rotation in the door with its extremely rigid and fitting design. D also scored highest in the force against the lock category. This means it would be very strong against any force exerted against it with its rigid and metal design. The design is especially strong in the first in a pocket category, with its flat and smaller design easily fitting in a pocket (with respect to our initial measurements of a pocket). It is also lightweight which contributes to our weight category as well as the pocket category. However, it is very weak in the occupation label category, with our initial design not having an obvious "occupied" label or anything of the sorts.

Potential Iterations:

- Occupied labels can easily be stickered onto the side of the metal, clearly and obviously showing that the stall is occupied.
- The rubber bands might potentially snap, so figuring out a more stable and reliable solution would be ideal.

Course of Action:

The due date of this project is realistically as soon as possible. Starting Feb 3rd we will be prototyping and finalizing our design which will take about two days. Testing and iterating will take about another three days.

Pivoting Designs:

Feb 6th,

After some quick initial testing of design D, we realized that it just didn't work the way we wanted it to. Our first mistake was grading the door swing criteria wrong. Our brief initial testing proved that the plates would just flip when rotating the door with any considerable push. This initial testing proved a fatal flaw within our design that could not be easily fixed without drastically changing the initial design beyond regular iterations. One of our teammates made a 3d printed prototype of Design B and after brief testing we realized that it was much more effective than originally believed to be. We quickly realized that it had very easy iterations we could make and we would like to pursue this design instead.

Iterations on design B:

- Screw Bolt added to keep stable
- Occupied sticker easily equipped to the side of the design.
- Design sized down